

The project should be performed by 6-8 people at most..

Deadline of submission 5 days after final exam, please pay attention to notification of the course in case anything changes.
All the code should have a docstring for description of the code based on Google standard.

The language is Python, object oriented and based on the keras, pytorch or tensorflow.

You should upload your code and description in a notebook and slide for presentation. The report should be inside the notebook.

Upload through D2L:

1- Notebook(code, description and summarize report at the end of notebook)

2- Screenshot of notebook, or convert notebook to pdf file.

How to organize notebook:

1- Details of dataset:

the link and reference of database, apply data analysis methods , finding correlation for tabular data.

For images use plotting

2- Preprocessing steps: normalization, remove null values, outliers, ...

3- Model

3-1 you could train your model from scratch: you use batch normalization, drop out, ...

3-2 you could use pretrained models and fine tune it.

4- Hyper parameter tuning (Good to have it)

5- keep track of training loss and accuracy. And analyze your training and validation model.history or tensorboard..

6- Apply model.prediction

7- Apply model.evaluation

8- interpret your result.

9- Write your report. If you use GPUs, TPU or CPU, write details, write also about the memory.

10-Next step

(could be what is your suggestion if you want to continue on this project).

11- Lessons to learn. Each one of you in the team should have lessons to learn under your name and add it at the end of your presentation.

12- Presentation.

NOTE: All the students and groups should submit their work. All of you should be attached to your project. If you do not attach the project under your name you will lose your grade.

NOTE2: Don't forget to convert your notebook to a pdf file or image file and upload in as file of your project.

NOTE3: DO NOT ZIP the files. YOU MUST UPLOAD EACH FILE ONE BY ONE.